

Project Design Phase

Problem – Solution Fit Template

Date	09July 2026
Team ID	SWTID-2026-2791
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ Understand the existing situation in order to improve it for your target group.

Template:

Problem–Solution Fit Canvas			Project: Smart Agricultural Optimization Engine			CS	CC	AS
						Define CS, fit into CC		Explore AS, differentiate
1. CUSTOMER SEGMENT(S) CS Who is your customer? - Small and medium-scale farmers - Agricultural cooperatives - Agronomists and agricultural consultants - Government agriculture departments - Agricultural startups and agri-businesses		6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? - Limited budget and low purchasing power - Poor internet connectivity in rural areas - Low digital literacy - Limited access to IoT devices and sensors - Language barriers - Lack of trust in new technologies		5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? - Traditional farming based on experience (low accuracy) - Manual soil testing (time-consuming, costly) - Weather forecast from TV / mobile apps (general, not specific) - Advice from local experts / extension officers (not real-time) - Existing farm management apps (limited features, not integrated) Pros: Easily accessible, low cost Cons: Not personalized, lack real-time data, not predictive				
Focus on J&P, tap into BE, understand RC			Focus on J&P, tap into BE, understand RC					
2. JOBS-TO-BE-DONE / PROBLEMS J&P RC BE Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides. - Optimize irrigation and water usage - Predict crop yield accurately - Detect crop diseases and pest attacks early - Recommend optimal fertilizers and nutrients - Improve crop productivity and quality - Reduce farming costs and increase profitability - Provide real-time weather and soil insights - Support data-driven farming decisions		9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? - Lack of real-time monitoring of soil, weather and crop health - Unpredictable climate change and weather patterns - Inefficient use of water, fertilizers and resources - Limited access to actionable agricultural data and analytics - Delayed detection of diseases leading to crop loss - Fragmented and non-personalized information for farmers		7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? - Consult local agricultural experts - Check weather forecast on mobile / TV - Observe fields manually - Attend training programs and farm meetings - Use WhatsApp groups to seek advice - Visit seed and fertilizer stores for guidance				
Identify strong TR & EM			Extract online & offline CH of BE					
3. TRIGGERS TR What triggers customers to act? - Unpredictable weather conditions - Noticeable decline in crop yield - Pest or disease outbreak in crops - High input costs (water, fertilizers, seeds) - Government schemes and subsidies - Recommendations from peers or experts - Adoption of new farming technologies		8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7 - Mobile applications - WhatsApp groups - YouTube farming videos - Agricultural websites and portals - Social media (Facebook, Telegram groups) - SMS / Email alerts 8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development. - Visit agricultural extension offices - Attend local training and workshops - Village meetings and discussions - Visit seed and fertilizer stores - Agricultural exhibitions and events		10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour. Smart Agricultural Optimization Engine is an AI and IoT powered platform that analyzes real-time soil, weather and crop data to provide personalized recommendations for irrigation, fertilization, crop disease detection, yield prediction and resource optimization. It helps farmers make data-driven decisions, reduce costs, improve productivity and promote sustainable farming through an easy-to-use mobile application.				
4. EMOTIONS: BEFORE / AFTER EM Before - Worried about crop health - Uncertain about yield - Financial stress - Confused about decisions - Frustrated due to losses After - Confident in decision making - Higher productivity - Financially secure - Satisfied with crop performance - Peace of mind								
Purpose / Vision To empower farmers with intelligent, data-driven solutions that optimize agricultural resources, increase productivity, improve income, and promote sustainable and resilient farming.						TR	SL	CH

